

ROUNAK BURMAN

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PROFESSIONAL SUMMARY

Data Scientist and ML engineer with a Master's in Computer Science, specializing in computer vision, NLP, and LLM / agentic systems. Skilled across the full ML lifecycle — from YOLO-based detection and OCR document intelligence to multi-agent workflows built with LangGraph and A2A — and in shipping them with Docker, Kubernetes, and AWS, consistently delivering models exceeding 90% accuracy.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, HTML, CSS

ML & Computer Vision: PyTorch, TensorFlow, Scikit-Learn, OpenCV, YOLOv8, DeepSORT, Roboflow, pose estimation, keypoint detection

LLM & GenAI: LangGraph, A2A, multi-agent orchestration, RAG, embeddings, LLM agents, Whisper API, IndicParler TTS, BERT, Gensim, Mistral OCR, Docling, TwelveLabs, NLP

Data & Databases: PostgreSQL, MongoDB, Elasticsearch, SQL, Pandas, NumPy

Web & Cloud: React, MUI, Flask API, Firebase, Postman, AWS, Docker, Kubernetes

WORK EXPERIENCE

TCG Digital

Kolkata, India

Data Science Consultant

Jul 2025 – Present

- Architected a multi-agent saree-design tagging pipeline orchestrated with **LangGraph**, coordinating 5+ vision and LLM tools to extract visual features, auto-generate structured tags and embeddings, and persist design metadata to a MongoDB collection — cataloging 5,000+ saree designs and cutting manual tagging effort by 70%.
- Built the orchestrator to also generate new saree designs, answer sales-data queries, generate visualizations, run trend research with illustrated PPT and PDF outputs, and pick out and display catalogue designs linked to sales data via design ID.
- Built a document-intelligence pipeline with **Docling**, applying OCR and vision models to make PDFs and scanned PDFs LLM-ready with over 95% extraction accuracy.
- Integrated the **TwelveLabs** video API to auto-generate timestamps of sports events from live video feeds with over 85% accuracy.

Data Science Intern

Sep 2024 – Jun 2025

- Implemented computer vision and fine-tuned ML models for corrosion detection in metal pipelines with over 90% accuracy.
- Detected litter and spillage in real-time footage using DeepSORT and YOLO with over 91% accuracy.
- Translated soccer commentary from English to Hindi using the Whisper API and IndicParler TTS, and ran 3+ CV models (YOLOv8, YOLOv8x-pose) for pose estimation, keypoint detection, and object tracking.
- Applied a Google BERT embedding model with custom field-weighting to deduplicate patient records across two tables into a master table with 95% accuracy.
- Extracted and encoded pictogram data from two PDF versions and regenerated a merged PDF using Mistral OCR with over 98% accuracy.

University of Massachusetts, Boston

Boston, MA

Project Assistant, Full-Stack Development

Feb 2024 – May 2024

- Automated extraction and summarization of medical research articles in a web app using TensorFlow, Gensim, and BERT, increasing content accessibility for healthcare professionals by 25%.
- Developed a PyQt5 desktop app for labeling VAS data, improving labeling accuracy by 15%.

Lynchval Systems

Boston, MA

Database Engineer Intern

May 2023 – Jul 2023

- Enhanced predictive analytics by optimizing SQL database queries, reducing data interval times by 15%.

Curved Pixel LLP

Kolkata, India

Full-Stack Developer

Dec 2020 – Jun 2022

- Led development of 5+ dynamic web pages and 3 complex APIs using Python, React, JavaScript, HTML, CSS, MUI, and Flask, improving overall system efficiency and user engagement.
- Improved site response time by 20% through optimized JavaScript and CSS.

PROJECTS

Fletch Homes — Real Estate Web App

- Built a real estate web app with a team of 3 (5+ back-end and 7+ front-end components) using Flask, Firebase, Postman, AWS, and Docker.

iConcern

- Implemented two types of topic modeling and caching with NLP (Gensim, BERT) over a MongoDB-backed web app.

EDUCATION

University of Massachusetts, Boston

Boston, MA

Master of Science, Computer Science

May 2024

Kalyani University

West Bengal, India

Bachelor of Technology, Information Technology

Oct 2020